

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 1

Year: 2012

Subject: 392133 Physics III

Section: 5-6

Date: 28 September 2012

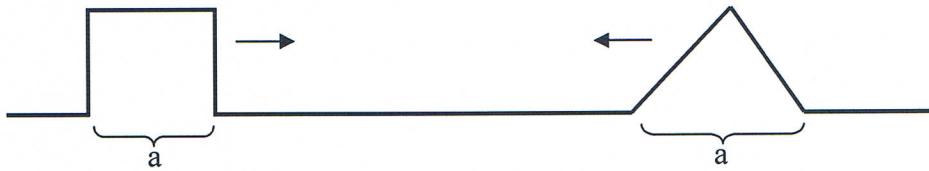
Time: 10.00-12.00

Name: _____ ID: _____ Field of Study: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Text books are NOT allowed.
4. Dictionaries and calculator are permitted.
5. No any electronic communication devices allow in the exam room.
6. The examination has 6 pages (including this page), 12 questions and a total score of 30 points.
7. Write all your answers on this exam sheet.

1. Two pulses move toward each other as illustrated. Sketch the resultant shape of the medium when the two pulses overlap. (1 point)



2. A laser beam is incident at an angle of 30.0° to the vertical onto a solution of corn syrup in water. If the beam is refracted to 19.24° to the vertical, (a) what is the index of refraction of the syrup solution? Suppose the light is red, with wavelength 632.8 nm in a vacuum. Find its (b) wavelength, (c) frequency, and (d) speed in the solution. (4 points)

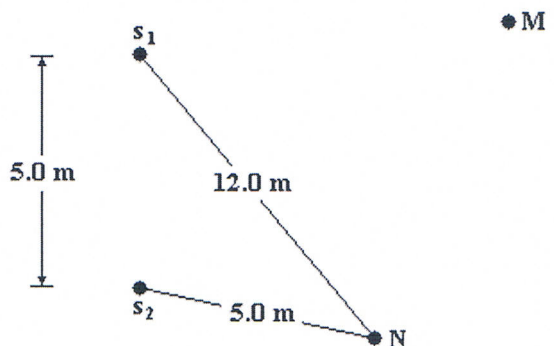
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3. Red light with a wavelength of 650 nm travels from air into glass ($n = 1.51$). (2 points)

- (a) What will its velocity be in the glass?
- (b) What will its wavelength be in the glass?

4. Two very small, identical loudspeakers, each of which is radiating sound uniformly in all directions, are placed at points S_1 and S_2 as shown in the diagram. They are connected to a source so that they radiate, in phase, a sound with a wavelength of 2.00 m. The speed of sound may be taken as 340 m/s. (3 points)

- (a) What is the frequency of the radiated sound?
- (b) Point M , a nodal point, is 7.0 m from S_1 and more than 7.0 m from S_2 . What are three possible distances M could be from S_2 ?



- (c) Point N , also a nodal point, is located 12.0 m from S_1 and 5.0 m from S_2 . On what nodal line is it located?

5. With two slits 0.12 mm apart, and a screen at a distance of 80 cm, the third bright line to one side of centre in an interference pattern is found to be displaced 9.0 mm from the central line. What was the wavelength of the light used? What colour was it? (2 points)

6. Helium-neon laser light ($\lambda = 6.328 \times 10^{-7}$ m) passes through a single slit with a width of 43 μm onto a screen 3.0 m away. What is the separation of adjacent minima, other than the central maximum? (2 points)

7. A man 1.80 m tall stands in front of a mirror and sees his full height, no more and no less. If his eyes are 0.14 m from the top of his head, what is the minimum height of the mirror?

(2 points)

8. An object is placed 20.0 cm from a concave spherical mirror having a focal length of magnitude 40.0 cm. (4 points)

- (a) Use graph paper to construct an accurate ray diagram for this situation.
- (b) From your ray diagram, determine the location of the image.
- (c) What is the magnification of the image?
- (d) Check your answers to parts (b) and (c) using the mirror equation.

9. A convex spherical mirror, whose focal length has a magnitude of 15.0 cm, is to form an image 10.0 cm behind the mirror. (a) Where should the object be placed? (b) What is the magnification of the mirror? (2 points)

10. Estimate how far away a cicada can be heard if the lowest audible intensity level of the sound it produces is $1.0 \times 10^{-12} \text{ W/m}^2$ and the power of a cicada's sound source is $2.0 \times 10^{-6} \text{ W}$.
(2 points)

11. Certain sunglasses use a polarizing material to reduce the intensity of light reflected from shiny surfaces, such as water or the hood of a car. What orientation of the transmission axis should the material have to be most effective? (2 points)
12. Two submarines are underwater and approaching each other head-on. Sub A has a speed of 12.0 m/s and sub B has a speed of 8.00 m/s. Sub A sends out a 1550 Hz sonar wave that travels at a speed of 1522 m/s. (a) What is the frequency detected by sub B? (b) Part of the sonar wave is reflected from B and returns to A. What frequency does A detect for this reflected wave? (4 points)

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